Young Woo Choi

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Education

2016–2021	Ph.D. in Physics, Department of Physics, Yonsei University, Seoul, Korea (Advisor: Prof. Hyoung Joon Choi)
2010–2016	B.S. in Physics, Department of Physics, Yonsei University, Seoul, Korea (Mandatory military service from 2012 to 2014)

Academic Experience

6th BerkeleyGW Workshop and Conference, Oakland, California, USA, 2019

School on electron-phonon physics from first principles, ICTP, Trieste, Italy, 2018

Honors & Awards

Global Ph.D. Fellowship, National Research Foundation, 2017-2021 (20,000 USD/year)

Yonsei Alumni Scholarship, 2012, 2014-2015 (10,000 USD/year)

Miraeasset Scholarship, 2011 (6,000 USD/year)

Excellent Paper Award, Korea Institute of Science and Technology Information, 2019

Excellent Presentation Award, The 15th KIAS Electronic Structure Calculation Workshop, 2019

Excellent Presentation Award, Korean Physical Society Spring Meeting, 2019.

Excellent Presentation Award, APCTP-KIAS Quantum Materials Symposium, 2019

Excellent Presentation Award, Korean Physical Society Spring Meeting, 2017

Best Presentation Award, KIAS Physics Winter Camp, 2016

Research Interests

- ***Ab initio* calculation of electronic and transport properties of low-dimensional materials**
  - Moiré superlattices, twisted graphene layers, 2D semiconductors, Fe-based superconductors
- **Computational study on many-body interactions in solids**
  - Electron-phonon interaction, electron correlation, GW, DFT+DMFT
- **Development of electronic-structure calculation methods using high-performance computing**
  - Massively parallel *ab initio* calculation, large-scale tight-binding calculation

Publications

1. **Strong electron-phonon coupling, electron-hole asymmetry, and nonadiabaticity in magic-angle twisted bilayer graphene,**
   Young Woo Choi and Hyoung Joon Choi,
   Physical Review B **98**, 241412(R) (2018). [arXiv:1809.08407] *cited 79 times*
2. **Intrinsic band gap and electrically tunable flat bands in twisted double bilayer graphene,**
   Young Woo Choi and Hyoung Joon Choi**,**
   Physical Review B **100**, 201402(R) (2019). [arXiv:1903.00852] *cited 27 times*
3. **Role of Electric Fields on Enhanced Electron Correlation in Surface-Doped FeSe,**
   Young Woo Choi and Hyoung Joon Choi,
   Physical Review Letters **122**, 046401 (2019). [arXiv:1901.04062] *cited 3 times*
4. **Anisotropic Pseudospin Tunneling in Two-Dimensional Black Phosphorus Junctions,**
   Young Woo Choi and Hyoung Joon Choi**,**
   (submitted)

5. **Atomistic Study on Electron-Phonon Coupling in Twisted Graphene Layers,**
   Young Woo Choi and Hyoung Joon Choi**,**
   (in preparation)
6. **Giant energy-gap two-dimensional topological insulator based on hexagonal network of carbon,**
   Woochang Kim, Young Woo Choi, and Hyoung Joon Choi,
   (in preparation)
7. **Experimental determination of valence electronic dispersion of orthorhombic $CH_3NH_3PbI_3$ studied by high-resolution angle-resolved photoelectron spectroscopy,**
   Jeehong Park, Young Woo Choi, Donghee Kang, Soonsang Huh, Changyoung Kim, Hyoung Joon Choi, Yeonjin Yi,
   (in preparation)

International Conference Presentations

1. **Atomistic study of electron-phonon coupling in magic-angle twisted bilayer graphene.**
   Contributed talk, American Physical Society March Meeting, Boston, MA, USA, March 4-8, 2019.
2. **Role of the charge-transfer induced electric field in potassium-dosed FeSe layers.**
   Contributed talk, American Physical Society March Meeting, Los Angeles, CA, USA, March 5-9, 2018.
3. **Effects of the potassium dosing on the electronic correlation in FeSe: DFT+DMFT study.**
   Contributed talk, American Physical Society March Meeting, New Orleans, LA, USA, Mar 13-17, 2017.
4. **Atomistic Study of Electronic Structure and Electron-Phonon Coupling in Twisted Graphene Layers.**
   Poster, The 22nd Asian Workshop on First-Principles Electronic Structure Calculations, Osaka, Japan, October 28-30, 2019
5. **Atomistic Study of Electronic and Phononic Properties of Twisted Graphene Layers.**
   Contributed talk, The 14th Asia Pacific Physics Conference, Kuching, Malaysia, November 17-21, 2019.

Domestic Conference Presentations

1. **Electron-Phonon Coupling in Twisted Double Bilayer Graphene.**
   Contributed talk, The Korean Physical Society Fall Meeting, Gwangju, Korea, October 23-25, 2019.
2. **Role of Electric Fields on Electron Correlation in Surface-Doped FeSe.**
   Poster, 11th International Conference on Magnetic and Superconducting Materials, Seoul, Korea, August 17-24, 2019.
3. **Atomic and Electronic Structure of Twisted Graphene Layers.**
   Poster, The 15th KIAS Electronic Structure Calculation Workshop, Korea Institute for Advanced Study, Seoul, Korea, July 4-5, 2019.
4. **Atomistic Study on the Electronic Structure of Twisted Double Bilayer Graphene.**
   Contributed talk, The Korean Physical Society Spring Meeting, Daejeon, Korea, April 24-26, 2019.
5. **Effects of electric fields on electron correlation in surface-doped FeSe.**
   Poster, APCTP-KIAS Quantum Materials Symposium 2019, YongPyong, Korea, February 10-15, 2019.
6. **Atomistic study of electron-phonon interaction in magic-angle twisted bilayer graphene.**
   Poster, APCTP-KIAS Quantum Materials Symposium 2019, YongPyong, Korea, February 10-15, 2019.
7. **Structural relaxation, electronic structure, and electron-phonon coupling in magic-angle twisted bilayer graphene.**
   Poster, The 21st Asian Workshop on First-Principles Electronic Structure Calculations, KAIST, Daejeon, Korea, October 29-31, 2018.
8. **Electron-phonon interactions in magic-angle twisted bilayer graphene.**
   Contributed talk, The Korean Physical Society Fall Meeting, Changwon, Korea, October 24-26, 2018.
9. **First-principles investigation of off-diagonal electron-phonon interaction in graphene.**
   Contributed talk, Korean Physical Society Spring Meeting, Daejeon, Korea, April 25-27, 2018.

10. **First-principles study on the enhanced electronic correlation in the potassium-doped surface layer of FeSe.**
    Contributed talk, Korean Physical Society Spring Meeting, Daejeon, Korea, April 19-21, 2017.
11. **Doping dependent electronic correlations in FeSe: DFT+DMFT study.**
    Poster, The Korean Physical Society Fall Meeting, Gwangju, South Korea, October 19-21, 2016.
12. **First-principles study of subsurface defects in topological insulator Bi_2Se_3 including van der Waals interaction.**
    Poster, Korean Physical Society Spring Meeting, Daejeon, Korea, April 20-22, 2016.

Computational Skills

***Ab initio* electronic structure calculation**
– SIESTA, Quantum ESPRESSO, EPW, BerkeleyGW, WIEN2k+EDMFT
Numerical Computing
– Experienced: Fortran, Python, MPI, Mathematica / – Basic: C++, Julia
General Computing
– Linux cluster operation & maintenance / – Web programming: PHP, MySQL, HTML, CSS, JavaScript

Teaching Experience

TA, Quantum Mechanics (Graduate Course), Spring 2016, Fall 2016, Spring 2017, Spring 2018, Fall 2018
TA, Statistical Mechanics (Undergraduate Course), Fall 2019, Fall 2020
TA, Thermal Physics (Undergraduate Course), Spring 2020
TA, Introduction to medical physics (Graduate Course), Fall 2017

Reference

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